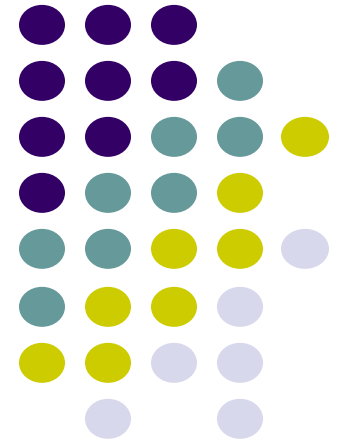


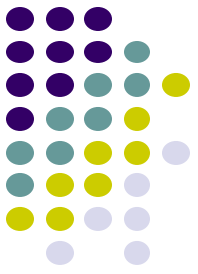
ErSE222: Machine learning in Geoscience

March. 12th, 2025

Tariq Alkhalifah and Omar Saad



Skip connections



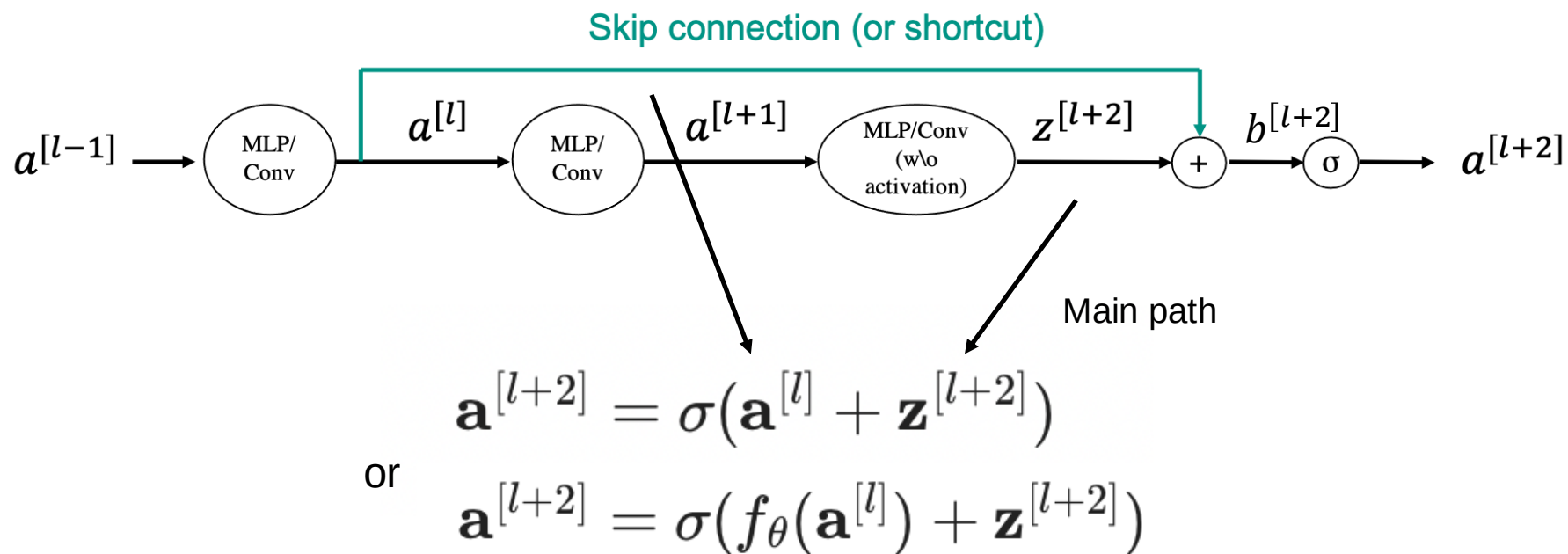
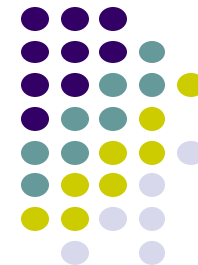
Deep neural networks can learn more complex relationships...

... but they are prone to slow training due to **vanishing gradients**

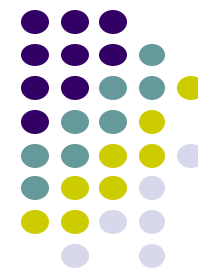
Skip connections: introduced in 2015 under the name of **Residual blocks**

take the activation of a certain layer and feed it directly to another layer further down in the computational graph.

Skip connections



Skip connections



Why successful?

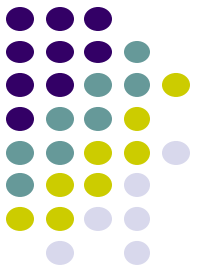
$$\mathbf{b}^{[l+2]} = \mathbf{a}^{[l]} + \mathbf{z}^{[l+2]} = \mathbf{a}^{[l]} + f_{l+2}(f_{l+1}(\mathbf{a}^{[l]}))$$

$$\mathbf{a}^{[l+2]} = \sigma(\mathbf{b}^{[l+2]})$$

$$\frac{\partial \mathbf{b}^{[l+2]}}{\partial \mathbf{a}^{[l]}} = 1 + \frac{\partial \mathbf{a}^{[l+1]}}{\partial \mathbf{a}^{[l]}} \frac{\partial \mathbf{z}^{[l+2]}}{\partial \mathbf{a}^{[l+1]}}$$

ensures that gradient is never 0!

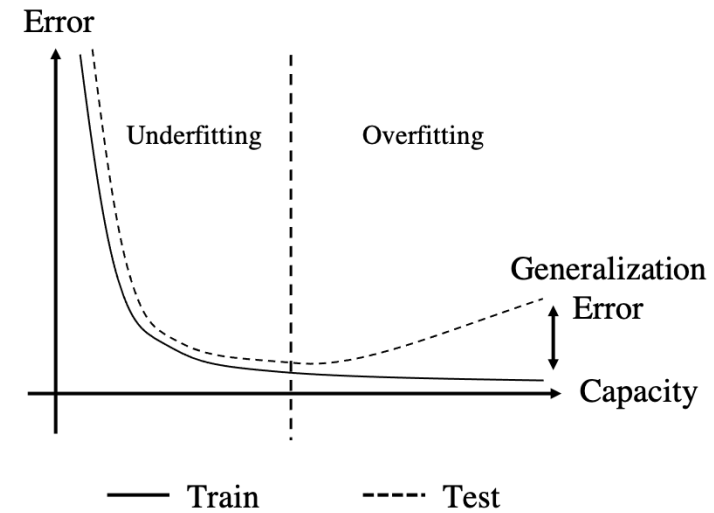
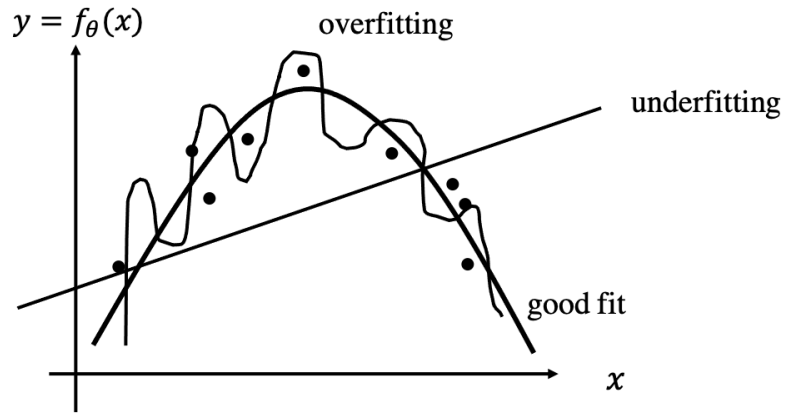
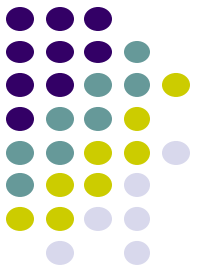
ML performance



Two different (usually contrasting) goals:

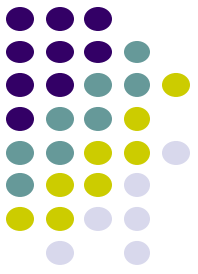
- **Bias:** ability to produce a small training error. When the error is small, we say that we have a model with low bias. Vice versa, when the error is large, the model have high bias. In this case, the model is said to be *underfitting* the data;
- **Variance:** ability to produce comparable training and testing error. In other words, if we define *gap* to be the difference between the testing and training errors, this is also the ability to produce a small gap. When the gap is large, the model is said to be *overfitting* the data.

ML performance

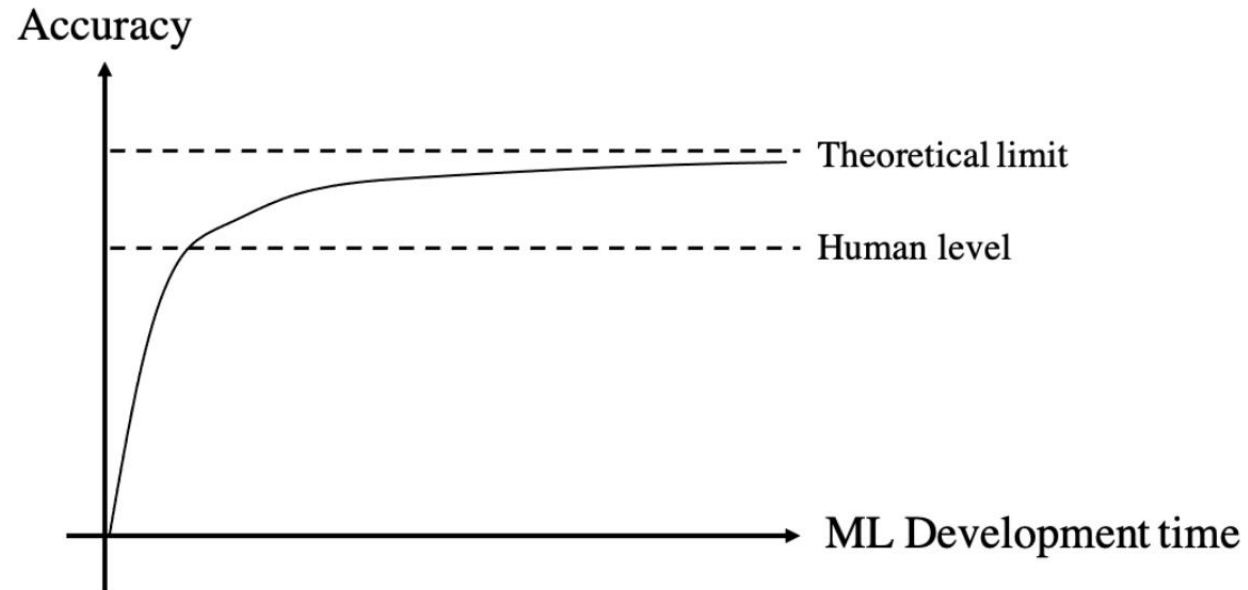


Capacity = size of the model

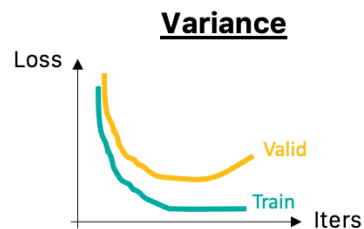
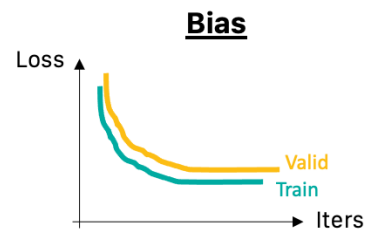
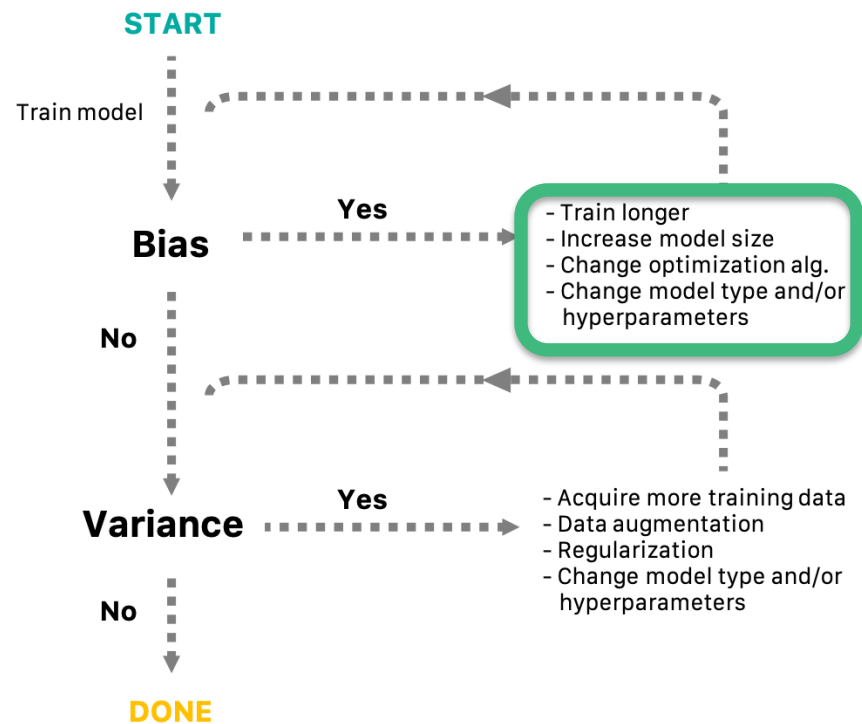
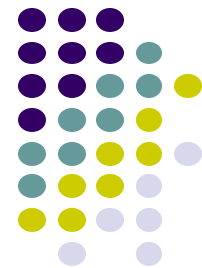
Expectations from ML model



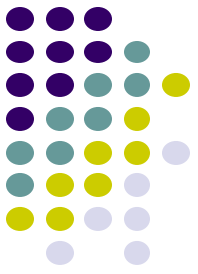
Always set expectations ahead of training!



Summary



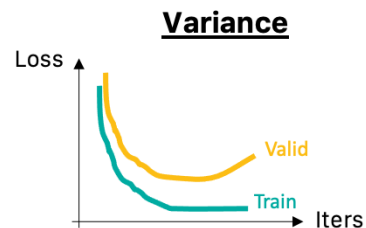
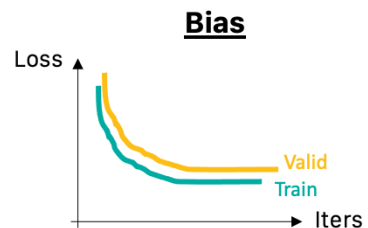
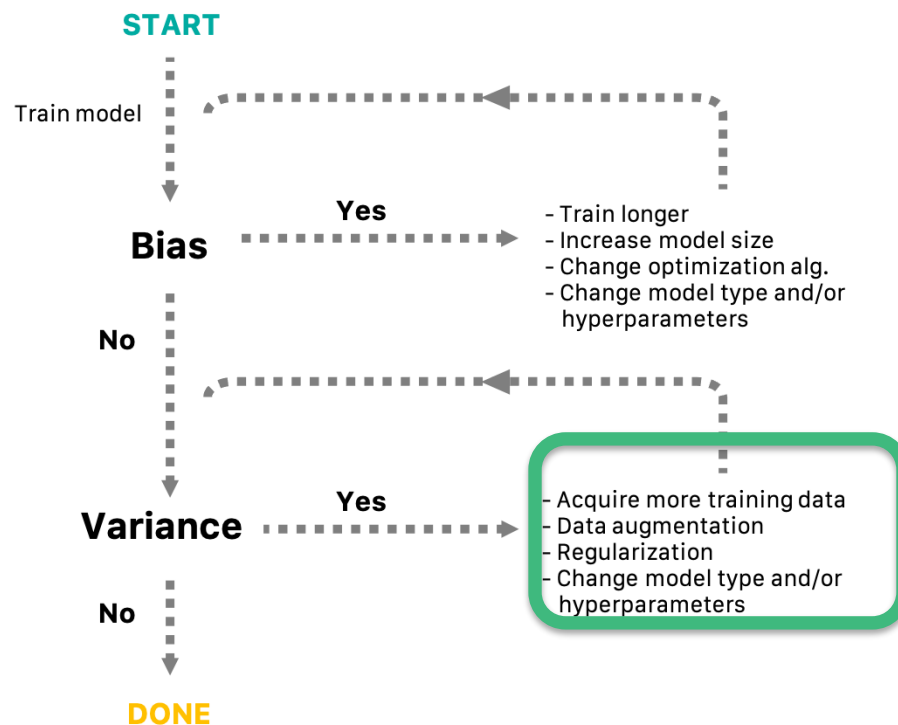
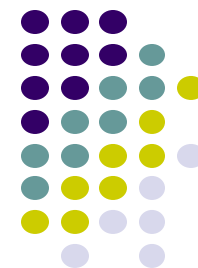
ML hyperparameters



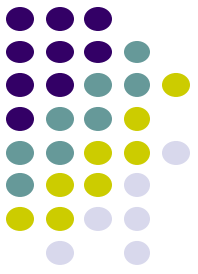
Hyperparameter: *parameter of a learning algorithm that is **not learned**, but chosen upfront*

- model size (capacity)
- activation function;
- optimizer,
- learning rate
- additional optimizer hyperparameters,
- batch size...

Summary



Data augmentation



Availability of large set of trustworthy training data = key to success of ML

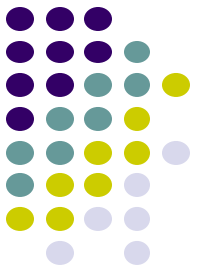
In the presence of few training data, apply transformations to enrich the dataset



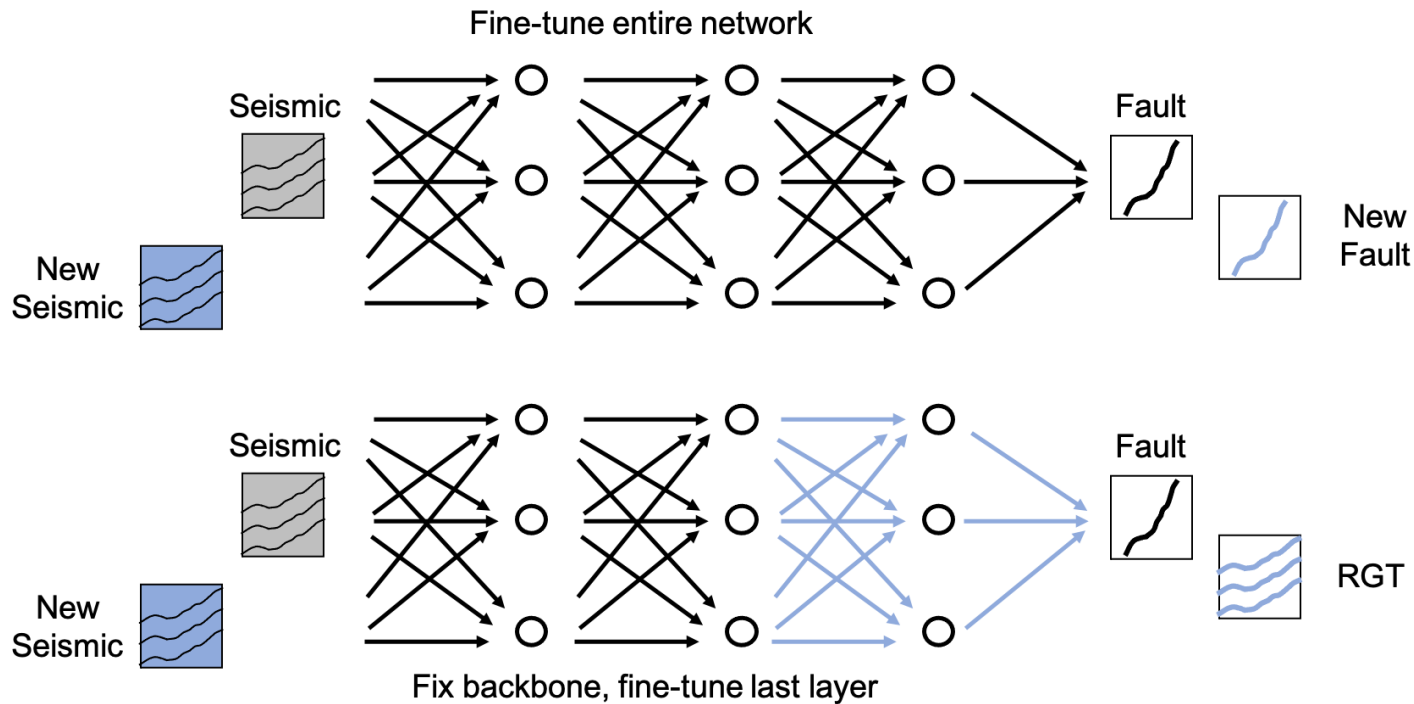
Ex: images

- cropping,
- flipping,
- rotating,
- shearing,
- averaging,
- color shifting

Transfer learning



Special case of data augmentation: use model trained in other data





Thank You

30 17:50